Theodoros Mangas

Backend Engineer — Python · Django · PostgreSQL

Komotini, Greece · Remote · open to EU/UK relocation · Employment or contract · theodoros-mangas.github.io/my-portfolio
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PROFILE

Python/Django engineer who builds, ships and runs web applications end to end: data modelling, REST APIs, Celery background jobs, automated testing and Docker-based deployment on Linux. I have delivered two production client platforms and built an engineering SaaS on my own, from first commit to live operation. A surveying and geoinformatics background adds depth in geospatial data and coordinate systems.

EXPERIENCE

Independent Python / Django Developer — Freelance & own products — 2022 – Present · Remote

- Build and maintain Django applications end to end for clients and my own products: data model, backend, interfaces, deployment and ongoing support.
- Operate production apps as Docker Compose stacks on a Hetzner Cloud VPS: Gunicorn behind a shared Nginx reverse proxy, Let's Encrypt TLS, loopback-only container ports, per-container memory limits, and written deploy, backup and restore runbooks.
- Git branches, pull requests, self-review and GitHub Actions CI; also Excel/VBA automations for local businesses.

SELECTED SOFTWARE DELIVERY

HandyTools — Engineering & utilities platform (own product) — Jul 2025 – Present · restricted beta

Django 6 · DRF · Celery · Redis · PostgreSQL 17 · Docker Compose · Nginx · Leaflet

- Built solo (230+ commits): PDF, image and OCR tools plus a full-screen Leaflet **Surveyor Toolkit** for Greek cadastral work, in English and Greek, with Free/Pro/Enterprise tiers enforced server-side via DRF permissions.
- Designed **Celery/Redis** pipelines for OCR and PDF conversion with bounded retries for transient I/O errors, late acknowledgement, time limits and atomic result writes. Uploads up to 50 MB are staged on disk and passed to workers by path, keeping them out of Redis task messages. Fixed a retry bug that discarded successful results and refunded free-tier quota, locked in by a regression test.
- **~700 Django tests** in GitHub Actions alongside ruff, mypy and a missing-migration check, with Docker images pushed to GHCR. Greek coordinate transformations (HEPOS, HATT) are tested to the millimetre against official values.
- Enforced per-user quotas under concurrent requests with **select_for_update** row locks, and removed N+1 queries in list views with select_related.

Iasmos Culture Club Museum — Paid client project — Delivered Sep 2026 · live · museum-iasmos-cultureclub.gr

Django 6 · SQLite · Docker · Gunicorn · Nginx · django-modeltranslation

- Bilingual (Greek/English) museum platform with a custom moderator CMS for exhibits, activities, events, pages and media; soft delete with superuser-only restore and purge.
- Privacy by design: self-hosted analytics with daily-rotating visitor hashes, consent-gated YouTube and Maps embeds, moderator Markdown sanitised with nh3, and django-axes login protection.
- **355 automated tests**; documented backup and restore procedures for the database and media.

Agropixida — Engineering practice platform — Live since 2022 · agropixida-syn2.gr

Django · DRF · MySQL · JavaScript · Leaflet · ezdxf · pyproj

- Bilingual business platform with service pages, a project showcase, and customer management with an audit trail (django-simple-history).
- Public toolbox of engineering utilities: DXF generation from coordinate files, and a vector converter between DXF, KML/KMZ and GeoJSON with ΕΓΣΑ87 ↔ WGS84 reprojection.

CityCare — Volunteer team coordination app — Django 5 · Bootstrap · GitHub Actions

- Teams log actions at mapped locations, with per-team daily statistics, rate-limited form posts and Greek/English UI; 35 tests run in CI.

TECHNICAL SKILLS

Backend & data	Python, Django, Django REST Framework, Celery, Redis, PostgreSQL, MySQL, SQLite; ORM optimisation, row locking, rate limiting, django-axes
Delivery & quality	Docker Compose, Linux, Nginx, Gunicorn, Git, GitHub Actions, Django test framework (unittest), ruff, mypy, Sentry
Geo & frontend	JavaScript, HTML/CSS, Bootstrap, Leaflet; pyproj, ezdxf, QGIS; DXF, KML/KMZ, GeoJSON, Shapefile

EARLIER & CONCURRENT EXPERIENCE

Land Surveyor & Geoinformatics Engineer — Independent practice, Greece — 2014 – Present · part-time

Surveying, CAD and GIS work within a multidisciplinary engineering collaboration, plus practical automation for data processing and validation.

EDUCATION, CERTIFICATION & LANGUAGES

B.Sc. Surveying & Geoinformatics Engineering — TEI of Central Macedonia, 2013

Advanced Python Developer Certificate — PeopleCert, 2022 · Greek (native) · English (fluent)